WORKED SOLUTION

STEP I 2001, Question 6

Pure • Surfaces of Revolution

STEP I 2001, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q6

This question has a full guided treatment in *Beyond STEP*, Section 4.8, *Slicing a Spherical Loaf*: https://beyondstep.keithmcnulty.org/ch04/spherical-loaf.html

The key idea

Generate the sphere by rotating the semicircle $y = \sqrt{a^2 - x^2}$ about the x-axis. In the surface-area formula the integrand $y\sqrt{1 + (\mathrm{d}y/\mathrm{d}x)^2}$ turns out to be *constant*, so the crust on any slice depends only on its thickness and not on where the slice is. For the second part, we eat slices from one end (as the hint's limits $-a$ to $-a + t$ suggest), so what is left is a spherical cap; the ratio V/A becomes a quadratic in the cap's thickness, and we find its largest value.

Solution

The crust is proportional to the number of slices

Let the loaf have radius a and centre at the origin, so its surface is obtained by rotating

$$y = \sqrt{a^2 - x^2}, \qquad -a \le x \le a,$$

about the x-axis. Differentiating, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{x}{\sqrt{a^2 - x^2}} = -\dfrac{x}{y}$ for $|x| < a$, so

$$y\left(1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2\right)^{\frac{1}{2}} = y\left(\frac{y^2 + x^2}{y^2}\right)^{\frac{1}{2}} = y \cdot \frac{a}{y} = a,$$

using $x^2 + y^2 = a^2$ and $y > 0$. Hence the curved surface area of the sphere between the planes $x = x_1$ and $x = x_2$ (with $-a \le x_1 < x_2 \le a$) is

$$2\pi \int_{x_1}^{x_2} a \, \mathrm{d}x = 2\pi a \, (x_2 - x_1).$$

(The integrand is undefined only at the end-points $x = \pm a$, where it is still bounded, so the integral is fine.)

Now suppose the loaf is cut into N slices, each of thickness $h = 2a/N$. Every slice, wherever it lies, carries crust of area exactly $2\pi ah$. So if n slices remain, in any positions at all, the crust remaining is

$$A = n \cdot 2\pi ah,$$

which is proportional to n, the number of slices remaining. This is what we had to show. (It's Archimedes' old result: a sphere and its circumscribing cylinder have the same area between any two parallel planes.)

The crusty condition

Slices are eaten from one end, so after some have gone the remaining bread is the cap from $x = -a$ to $x = -a + t$ for some t with $0 < t \leq 2a$. From the first part, $A = 2\pi at$. For the volume, substitute $u = x + a$, so that $y^2 = a^2 - x^2 = (a - x)(a + x) = (2a - u)u$:

$$V = \pi \int_{-a}^{-a+t} (a^2 - x^2)\,\mathrm{d}x = \pi \int_0^t (2au - u^2)\,\mathrm{d}u = \pi\Big(at^2 - \frac{t^3}{3}\Big).$$

Therefore

$$\frac{V}{A} = \frac{\pi t^2\big(a - \frac{t}{3}\big)}{2\pi at} = \frac{t}{2} - \frac{t^2}{6a} =: f(t).$$

Ask yourself The ratio depends on how much has been eaten. Which value of t do we need to worry about?

The condition must hold after *any* number of slices have been eaten, so we need the largest value of f. Completing the square,

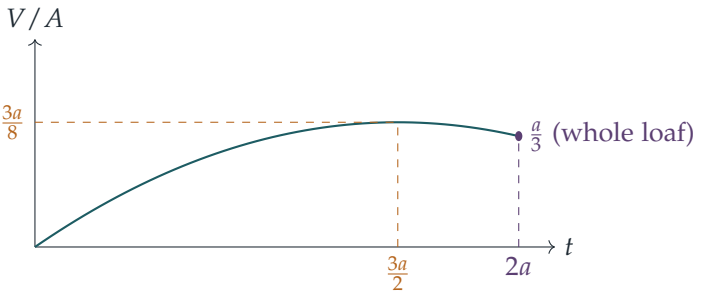
$$f(t) = \frac{3a}{8} - \frac{1}{6a}\Big(t - \frac{3a}{2}\Big)^2,$$

so f is largest when $t = \frac{3a}{2}$ (which lies in $(0, 2a]$), with maximum value $\frac{3a}{8}$.

Is $t = \frac{3}{2}a$ actually reached? It is if, for example, the loaf is cut into 4 slices and one is eaten (or into $4m$ slices and m are eaten). A loaf that is to be called crusty as a parallel-sliced loaf must satisfy the ruling for such a slicing, so it needs

$$\frac{3a}{8} < 1, \qquad \text{that is,} \qquad a < \frac{8}{3} = 2\tfrac{2}{3} \text{ metres},$$

as required. Conversely, if $a < \frac{8}{3}$ then $f(t) \leq \frac{3a}{8} < 1$ for every t, so the loaf is crusty however it is sliced.



The sketch shows $f(t)$ for $a = 2$: the ratio rises as we move towards the middle of the loaf and then falls, because near the far end we add a lot of crust but little bread.

Checks

- For the whole loaf, $t = 2a$ gives $f(2a) = a - \frac{2a}{3} = \frac{a}{3}$, which matches $\frac{4}{3}\pi a^3/(4\pi a^2)$.
- Take $a = \frac{8}{3}$ cut into 4 slices. With $1, 2, 3, 4$ slices left the ratio is $\frac{5}{9}, \frac{8}{9}, 1, \frac{8}{9}$, so after one slice is eaten the ratio is exactly 1, just failing the strict inequality.
- Numerically, quadrature of the area formula reproduced $2\pi a(x_2 - x_1)$ for random slices; and for $a = 2.66$ every slicing into up to 200 slices kept the ratio below 1.

Common slips

- Using only the whole loaf: $a/3 < 1$ gives $a < 3$, which is necessary but much weaker. The worst moment is when a quarter of the loaf's *width* has been eaten.
- Including the flat cut face in the crust. The crust is only the original curved surface, which is what the given formula measures.

Beyond STEP

The constant integrand is Archimedes' hat-box theorem: projecting a sphere sideways onto its circumscribing cylinder preserves area. That is the reason a point chosen uniformly on a sphere has a uniformly distributed height, a fact used constantly in probability and physics. Surface integrals like this one, and when maps preserve area, are developed in a first course on vector calculus; Needham, *Visual Complex Analysis*, has a beautiful geometric discussion of related maps of the sphere.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.